Listing 5-3. *State Change Example 2*

```
const int switchPin = 2;
const int ledPin = 13;

int state;
int lastState;

int buttonCounter = 0;

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(switchPin, INPUT);
}

void loop() {
  state = digitalRead(switchPin);
  if (state != lastState) {
    if (state == HIGH) {
      buttonCounter++;
    }
    lastState = state;
  }
  if (buttonCounter % 5 == 0) {
    digitalWrite(ledPin, HIGH);
    delay(20);
  } else digitalWrite(ledPin, LOW);
}
```

In this example, we have replaced the `ledState` variable with a `buttonCounter` variable to count the button presses. Instead of toggling the LED state, we are adding up each button press in this `if` statement:

```
if (state == HIGH) {
  buttonCounter++;
}
```

This will add one to the variable `buttonCounter` each time the input pin goes high. Then we need a way to test to see if we have hit five presses or not.

```
if (buttonCounter % 5 == 0) {
    digitalWrite(ledPin, HIGH);
    delay(20);
  } else digitalWrite(ledPin, LOW);
```

This final conditional statement does just that, by testing if the remainder of `buttonCounter` divided by 5 is equal to 0. If so, we know that the button has been pressed five times and the Arduino can turn on the LED. For any other number, the LED will remain off.